

Testing and Demonstration Report

Executive Summary: This week, we implemented and verified the STC89C52 simulator's core pipeline (FETCH → DECODE → EXECUTE), memory/stack routines, and custom FIFO queue operations (ENQ/DEQ). Both automated component unit tests and end-to-end assembly program simulations in the GUI executed with a 100% pass rate. System registers, stack pointer boundaries, and queue states matched expected values across all test cycles.

Demo Program and Execution: To demonstrate the pipeline and stack/queue workflow in action, and put together a test assembly program that performs sequential enqueues, dequeues, stack pushes, and pops

```
MOV A, #10 ; Load 0x0A into ACC
ENQ A ; Enqueue 0x0A into FIFO Queue
MOV A, #20 ; Load 0x14 into ACC
ENQ A ; Enqueue 0x14 into FIFO Queue
MOV A, #30 ; Load 0x1E into ACC
ENQ A ; Enqueue 0x1E into FIFO Queue
DEQ ; Dequeue front element (0x0A) into ACC
PUSH A ; Push ACC (0x0A) onto Stack
DEQ ; Dequeue next element (0x14) into ACC
PUSH A ; Push ACC (0x14) onto Stack
POP B ; Pop top of stack (0x14) into Register B
POP A ; Pop top of stack (0x0A) into Register A
END ; Halt Simulator
```

Line by Line Execution: During the step-by-step UI execution, we monitored the register values, stack pointer movements, and queue state at each clock cycle to verify state transitions

- ✓ **Step 1 (0x0000): MOV A, #10** — ACC: 0x0A | B: 0x00 | SP: 0x07 | Queue: []
- ✓ **Step 2 (0x0001): ENQ A** — ACC: 0x0A | B: 0x00 | SP: 0x07 | Queue: [0x0A]
- ✓ **Step 3 (0x0002): MOV A, #20** — ACC: 0x14 | B: 0x00 | SP: 0x07 | Queue: [0x0A]
- ✓ **Step 4 (0x0003): ENQ A** — ACC: 0x14 | B: 0x00 | SP: 0x07 | Queue: [0x0A, 0x14]

- ✓ **Step 5 (0x0004): MOV A, #30** — ACC: 0x1E | B: 0x00 | SP: 0x07 | Queue: [0x0A, 0x14]
- ✓ **Step 6 (0x0005): ENQ A** — ACC: 0x1E | B: 0x00 | SP: 0x07 | Queue: [0x0A, 0x14, 0x1E]
- ✓ **Step 7 (0x0006): DEQ** — ACC: 0x0A | B: 0x00 | SP: 0x07 | Queue: [0x14, 0x1E]
- ✓ **Step 8 (0x0007): PUSH A** — ACC: 0x0A | B: 0x00 | SP: 0x08 | Queue: [0x14, 0x1E] | RAM[0x08]: 0x0A
- ✓ **Step 9 (0x0008): DEQ** — ACC: 0x14 | B: 0x00 | SP: 0x08 | Queue: [0x1E] | RAM[0x08]: 0x0A
- ✓ **Step 10 (0x0009): PUSH A** — ACC: 0x14 | B: 0x00 | SP: 0x09 | Queue: [0x1E] | RAM[0x09]: 0x14
- ✓ **Step 11 (0x000A): POP B** — ACC: 0x14 | B: 0x14 | SP: 0x08 | Queue: [0x1E] | RAM[0x09]: 0x14
- ✓ **Step 12 (0x000B): POP A** — ACC: 0x0A | B: 0x14 | SP: 0x07 | Queue: [0x1E] | RAM[0x09]: 0x14
- ✓ **Step 13 (0x000C): END** — ACC: 0x0A | B: 0x14 | SP: 0x07 | Queue: [0x1E] | Program Halted

Final System State: When the program halted at 0x000C, the CPU state matched our manual calculations

PC: 0x000C

ACC: 0x0A

Register B: 0x14

SP: Restored to 0x07

FIFO Queue: [0x1E] (retained in buffer)

Data RAM: 0x0A at address 0x08, 0x14 at address 0x09

Test Verification

[PASS] DataMemory Read & Write

[PASS] Stack Push & Pop Operations

[PASS] FIFO Queue Enqueue & Dequeue

[PASS] CPU Initial State

[PASS] Full Program Execution Workflow

Test Summary: 5/5 Passed